

Produced-Water Lithium Extraction

Rezaee-calibrated DES/TOPO workflow for a pretreated
Smackover Li/Na stream

LI FEED

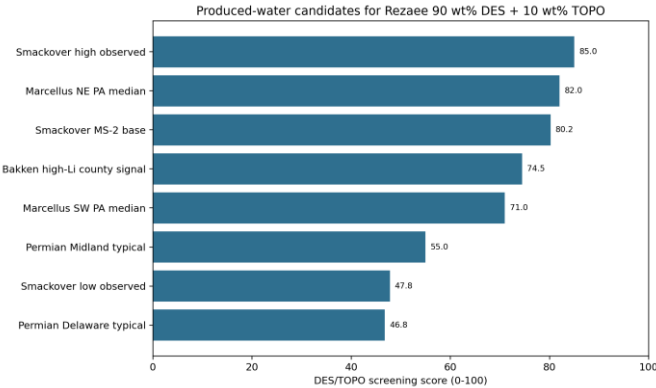
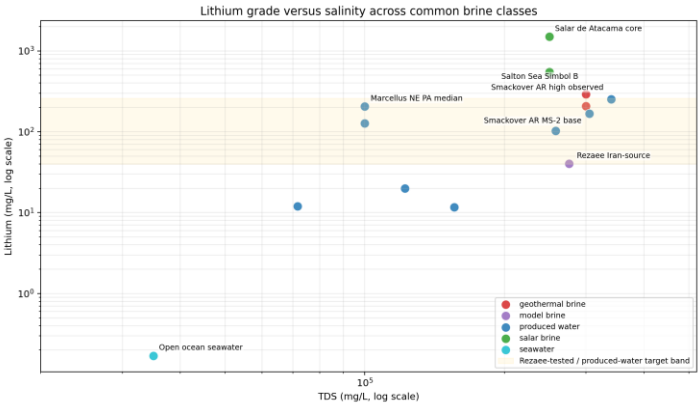
168 mg/L

NA FEED

64,100 mg/L

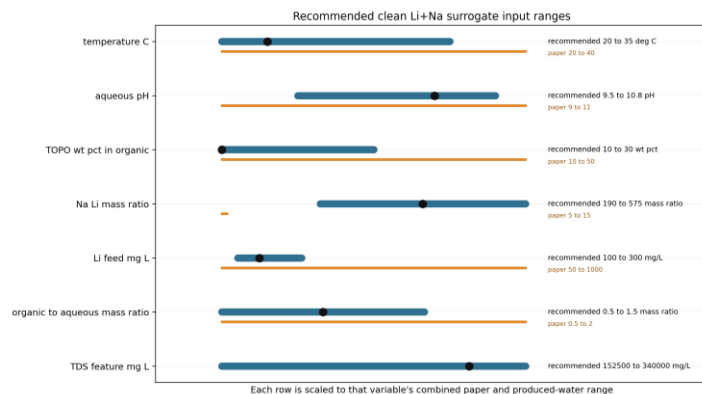
NOMINAL LI EXTRACTION

50.66%



Case Basis: Pretreated Smackover Li/Na Feed

The solvent model sees a clean Li/Na stream, not raw multication produced water.



Boundary condition: raw multication brine is never treated as the Rezaee extraction feed.

ENGINEERING READOUT

- 01 Begin extraction after divalent pretreatment; Ca/Mg/Sr/Ba stay outside the current solvent model.
- 02 Use the MS-2 clean feed as the nominal handoff row: 168 mg/L Li and 64,100 mg/L Na.
- 03 Retain TDS as the severity variable for process screening and uncertainty sweeps.

Rezaee DES/TOPO Surrogate Construction

The active model is a distribution surrogate tied to literature extraction evidence.



Model basis: calibrated_rezaee_section32_doe_basis_surface_distribution_surrogate
Caveat carried per row: outside original Rezaee Na/Li DOE domain

$$D_i = (E_i / (100 - E_i)) / (O/A)$$

ENGINEERING READOUT

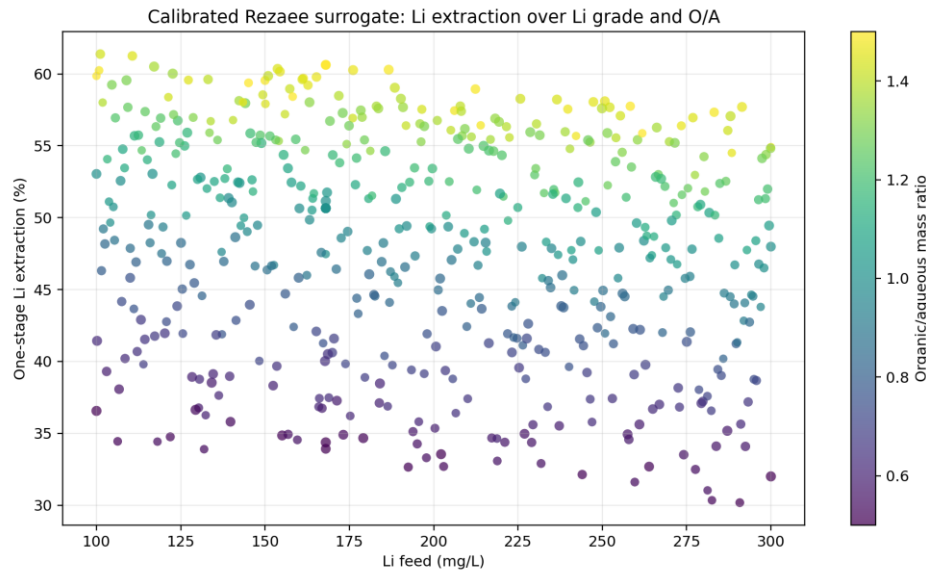
- 01 Convert Rezaee extraction responses into distribution coefficients using the organic-to-aqueous basis.
- 02 Fit log-distribution response surfaces from the calibrated Section 3.2 evidence table.
- 03 Evaluate 523 produced-water screening rows with model-basis and validity flags carried downstream.

Lithium Extraction Surface

The calibrated surrogate returns finite responses across the produced-water screening matrix.

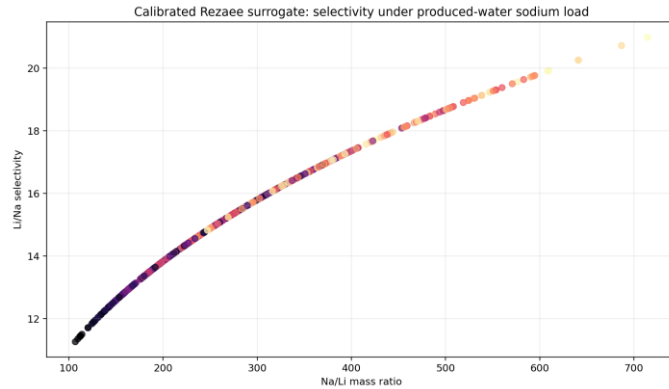
ENGINEERING READOUT

- 01 Lithium extraction spans 30.2-61.4% across the current screening matrix.
- 02 Nominal MS-2 one-stage lithium extraction is 50.66%.
- 03 The organic/aqueous ratio is the dominant process lever in the three-variable sweep.



Selectivity and Transfer Variables

The result is a staged-contacting contract, not only a recovery percentage.



Nominal MS-2 transfer variables

$$D_i = (E_i / (100 - E_i)) / (O/A)$$

E_{Li}	E_{Na}	$S_{\text{Li/Na}}$	D_{Li}	D_{Na}
50.66%	5.67%	17.09	1.0267	0.0601

O/A = 1.0, Li = 168 mg/L, Na = 64,100 mg/L

ENGINEERING READOUT

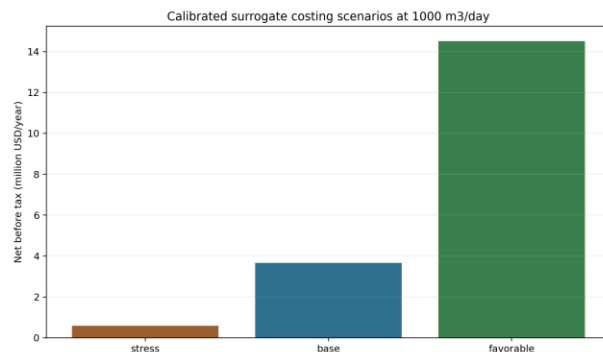
- 01 Nominal Li/Na selectivity is 17.09 at the clean MS-2 point.
- 02 Corrected distribution coefficients are the primary variables for MSContactor handoff.
- 03 Selectivity remains favorable under produced-water sodium loading, but the point is extrapolated beyond the original DOE range.

PrOMMiS/IDAES Handoff

Transfer variables are connected to staged recovery and screening economics.



Boundary condition: raw multication brine is never treated as the Rezaee extraction feed.



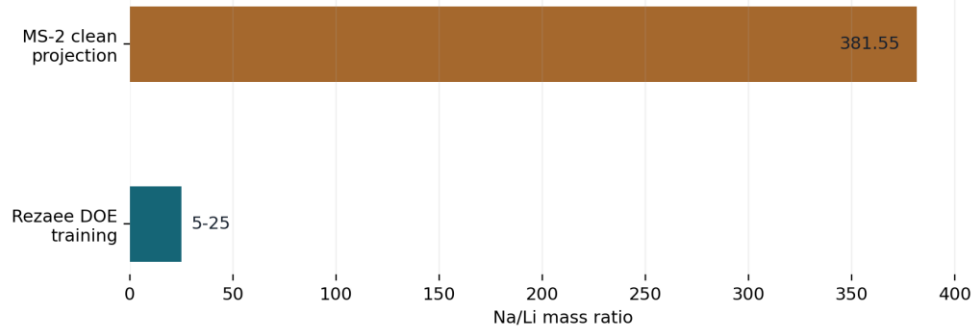
ENGINEERING READOUT

- 01 The handoff table carries extraction, selectivity, distribution coefficients, raffinate, and extract loading.
- 02 Costing scenarios use calibrated surrogate recovery values at the same feed-flow basis.
- 03 Economics stay screening-level until residence time, solvent loss, reagent demand, and product pricing are specified.

Validation Boundary

The key caveat is sodium-load extrapolation from Rezaee DOE space to Smackover water.

Extrapolation boundary carried into every surrogate row



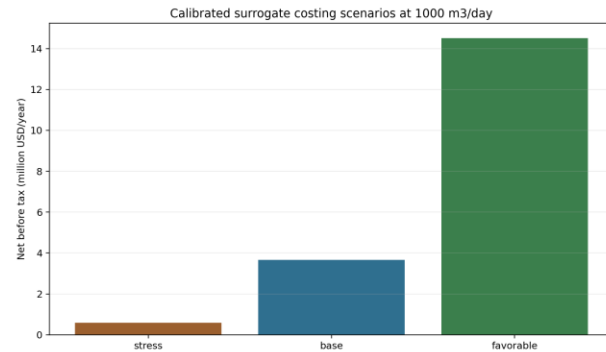
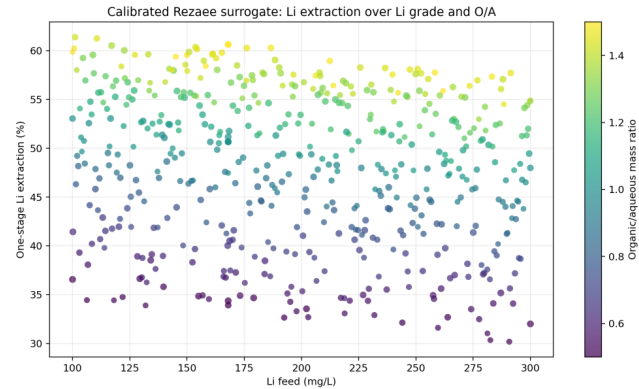
$$S_{\text{Li/Na}} = D_{\text{Li}}/D_{\text{Na}}$$

ENGINEERING READOUT

- 01 Every produced-water row is flagged as outside the Rezaee DOE Na/Li training domain.
- 02 The flag limits interpretation: process screening is appropriate; high-Na/Li validation remains future work.
- 03 Direct reactive-LLE closure is not claimed; the calibrated distribution surrogate is the presentation basis.

Presentation-Ready Chain

The current story now runs from feed definition through process-screening economics.



ENGINEERING READOUT

- 01 Use the calibrated Rezaee surrogate matrix as the active data product for the case-study deck.
- 02 Keep archived solvent-selection material outside the current Rezaee-calibrated presentation chain.
- 03 Next technical step: replace algebraic process handoff with a full ProMMiS MSContactor solve and costing block.